CCIT Report No. 383

On Joint Source-Channel Coding for the Wyner–Ziv Source and the Gel'fand–Pinsker Channel

Neri Merhav and Shlomo Shamai (Shitz)

ABSTRACT

We consider the problem of lossy joint source--channel coding in a communication system where the encoder has access to channel state information (CSI) and the decoder has access to side information that is correlated to the source. This configuration combines the Wyner—Ziv model of pure lossy source coding with side information at the decoder and the Shannon/Gel'fand—Pinsker model of pure channel coding with CSI at the encoder. We prove a separation theorem for this communication system, which asserts that there is no loss in asymptotic optimality in applying first, an optimal Wyner-Ziv source code and then, an optimal Gel'fand--Pinsker channel code. We then derive conditions for the optimality of a symbol--by--symbol (scalar) source--channel code, and demonstrate situations where these conditions are met. Finally, we discuss a few practical applications, including of overlaid communication where the model under discussion is useful.

Index Terms: Wyner-Ziv coding, Gel'fand–Pinsker coding, side information, channel state information, separation theorem, joint source–channel coding.